Space Instrument Testing Chamber (SITH): An Overview

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Abstract

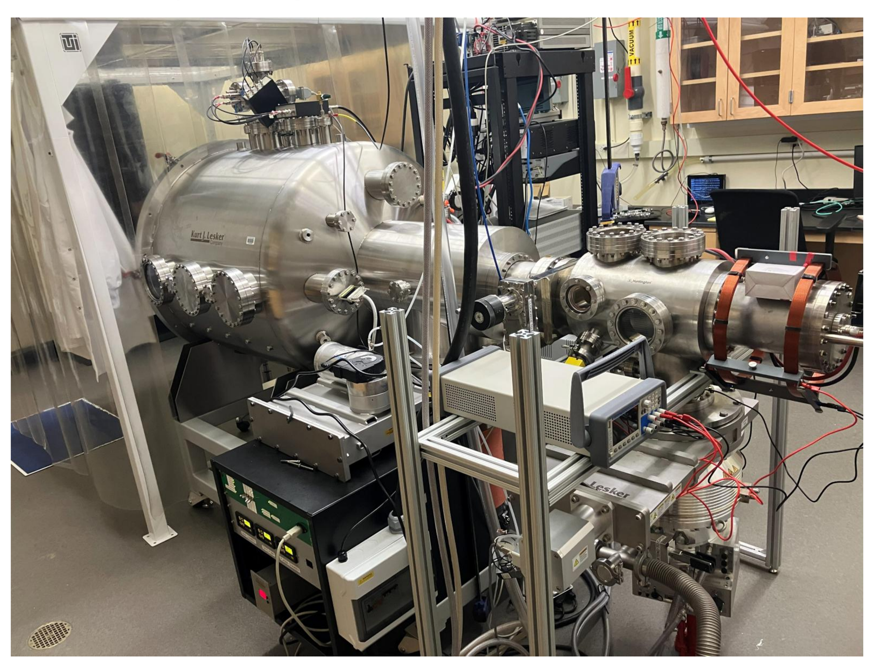
Getting to space is costly, time-intensive, and resource intensive. Therefore, it is important to fully calibrate scientific instruments before sending them into space. The Space Instrument Testing Chamber (SITH) is built to mimic the conditions of outer space to calibrate plasma and electric field measuring instruments. To calibrate electron plasma instruments, SITH uses a variable energy electron gun to mimic the differing energies the plasma instrument would observe in space. A Kimball physics electron gun is installed in the vacuum chamber. The vacuum chamber is pumped down to space relevant pressures of less than 8 x 10-9 Torr with a combination of turbo and cryo pumps. Inside the vacuum chamber is a 6-axis motion stage to allow control over detector position relative to the electron beam. SITH also includes an imaging microchannel plate (MCP) detector that provides measurements of the electron beam profile and count rates in specific regions. With this characterization, a known beam energy and flux is directed into an instrument under calibration. One of the first instruments to be tested within SITH is a proprietary solar blind particle detector. Here we present initial testing of such a detector at energies as low as 1 keV.

Introduction

Using a combination of a turbo vacuum pump and cryo pump, SITH maintains a pressure of 8 x 10^{-9} Torr

A 6-Axis motion stage on which instruments are mounted allows complete control over the orientation of the instrument with respect to the calibration electron beam.

A plasma chamber separated by a gate valve enables the pumps to keep both chambers under pressure while simultaneously testing plasma probes and space instruments

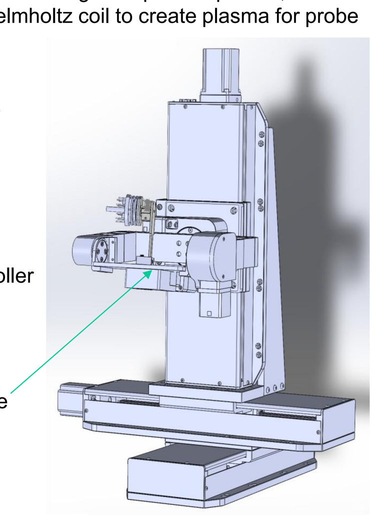


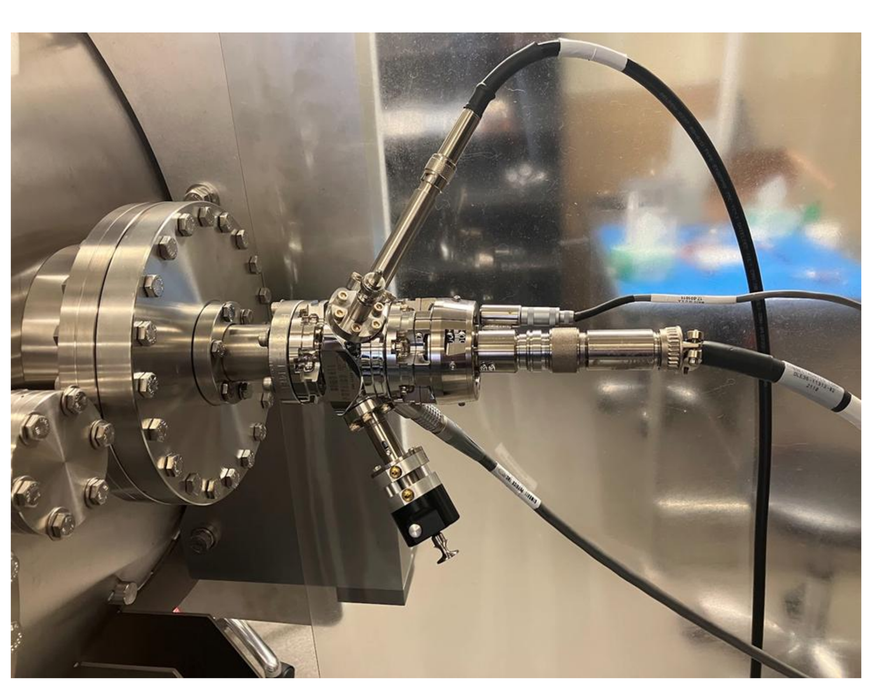
Space Instrument Testing Chamber (full view); main hatch in clean room to the left for ease of switching instruments. Plasma chamber on the right for plasma probes; uses argon with a filament source in tandem with a Helmholtz coil to create plasma for probe testing.

Custom, 6-Axis motion stage from Newmark Systems. Has 300mm of motion in the x,y,z planes, as well as rotational motion about each axis

Controlled via python script utilizing the Galil motor controller communications library

Mounting screws located here

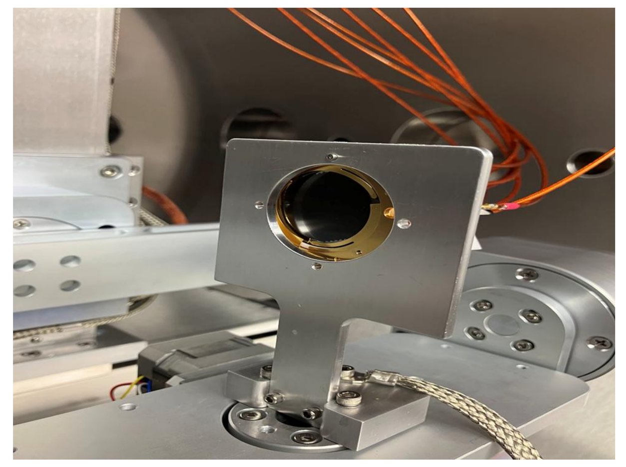




Kimball Physics EGG-3101 Electron Gun, capable of energies from 100-10000 eV, in spot sizes from 0.5-25 mm. Mounted horizontally, in-plane with the motion stage

Instruments and Apparatus

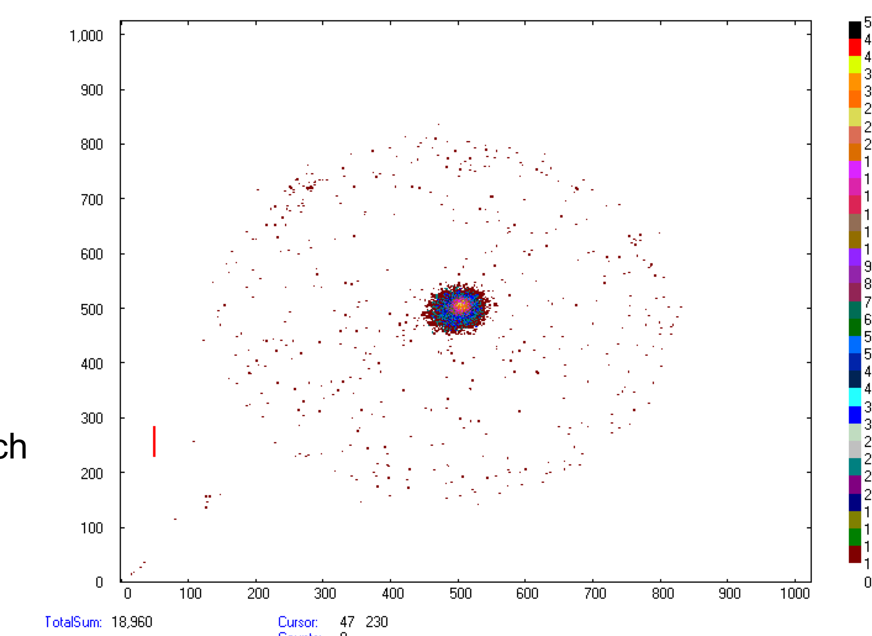
A 3-stack, imaging microchannel plate (MCP) detector measures the number of electrons that hit the plate, as well as their position on the plate. This produces a visual map of the electron beam profile and an absolute flux measurement.



MCP with a custom mount set up on the 6-Axis stage; lined up with the electron gun

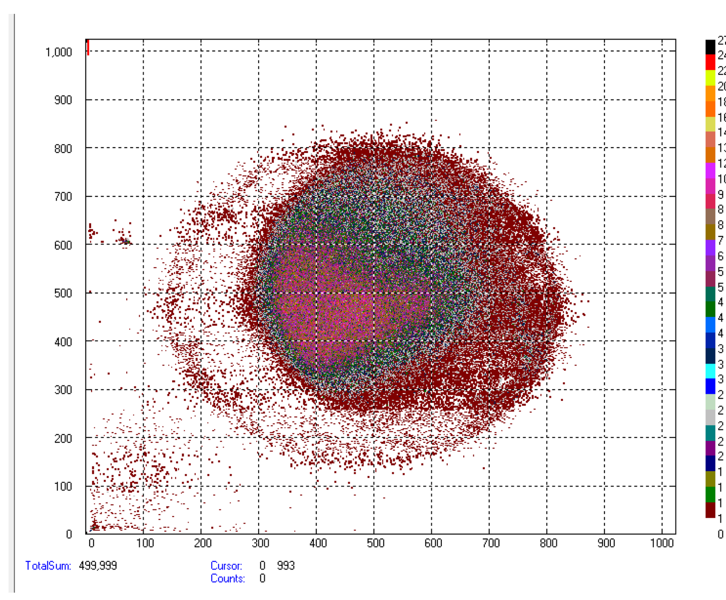
Mount is designed to be quickly swapped out with others to allow ease of switching detectors

MCP has an accompanying MPANT imaging software that that provides a spatial plot of the MCP surface and the electron counts associated with each pixel.



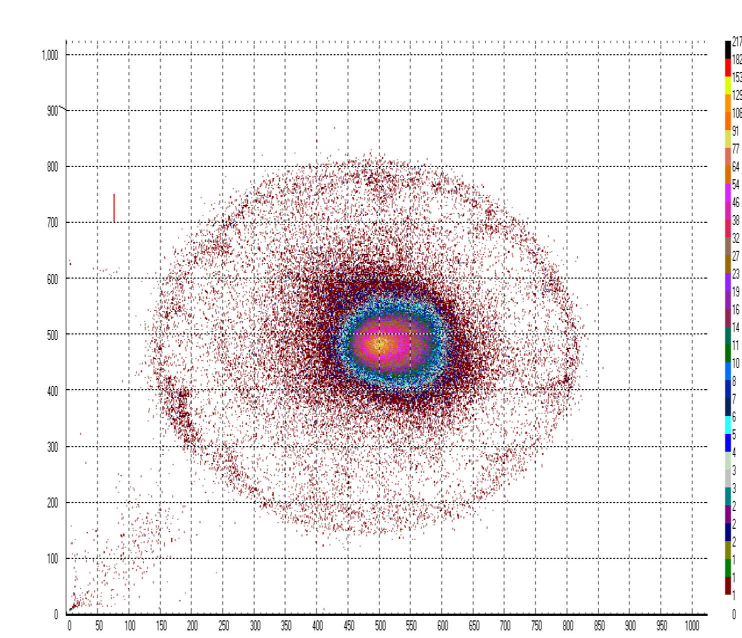
Shown above and below are plots of the electron beam with varying settings:

- The electron energy is varied between 0.1-10 keV.
- The beam current is varied to increase count rate
- The focus voltage and grid voltage are adjusted to alter to spot size between 0.5 and 25 mm. This has notable dependence on the working distance and beam energy.



By lowering the focus/energy ratio, the beam is spread across a larger area, but it makes it more challenging to maintain a uniform flux.

By lowering the grid voltage but maintaining the focus/energy ratio, the beam is over a larger region without losing its Gaussian intensity profile.

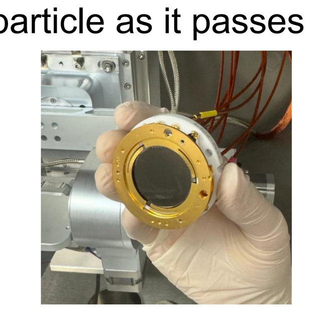


Why a Solid-State Detector

Low-energy plasma instruments typically employ one of two types of particle detectors: MCPs and SSDs.

Microchannel plates (MCPs): electron-multiplying detector made of a thin plate containing millions of microscopic channels (tiny glass tubes), each typically 10–20 µm in diameter.

Solid-State Detectors (SSDs): semiconductor-based particle detector, typically made from silicon, that measures energy deposited by a charged particle as it passes through the material.



Implementation of these detectors in low-energy plasma measurements typically follows one of two approaches:

- MCPs with complex electrostatic geometry and blackened surfaces to prevent detection of visible and UV wavelengths while still allowing ions and electrons to pass through.
- SSDs with an electrostatic analyzer and post-acceleration combo that only allows particles with the right (E/q) ratio to pass through without abandoning strong measurable signals. The SSDs typically have thick coatings to block UV photons. This raises the energy threshold of the SSD.

MCPs have their advantages, but they also have their fair share of challenges:

- Their gain degrades with time.
- They require careful outgassing before use.
- They require bias voltages of 3 – 5 keV.

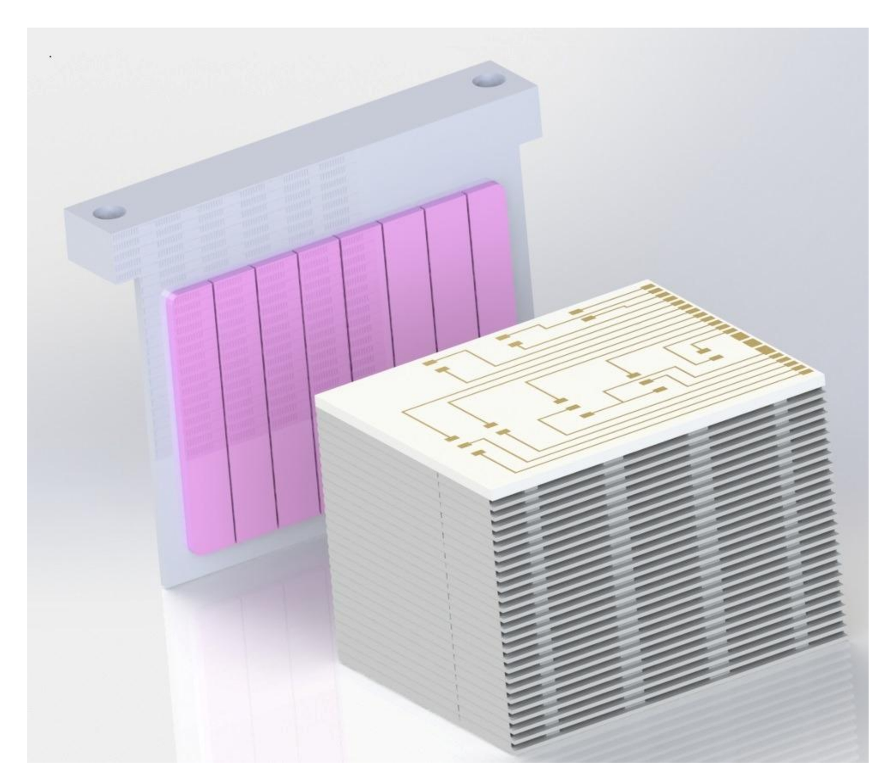
The challenges for SSDs include

- They have high energy thresholds for detection.
- Because of the high energy detection threshold, they require a high-energy post-acceleration (around 30 kV), which is challenging for a small space instrument.

For particle detectors with intrinsically less sensitivity to light, the focus has turned to **carbon-based solid-state detectors**. These detectors are a SSD made from synthetic CVD-grown carbon-based material instead of silicon. The advantages for this new solid-state detector include:

- The bandgap of the material is 5.47 eV, making it insensitive to wavelengths longer than 226 nm. The bulk of the solar emission spectrum is larger than 250 nm.
- Less sensitive to solar light than SSDs and MCPs.

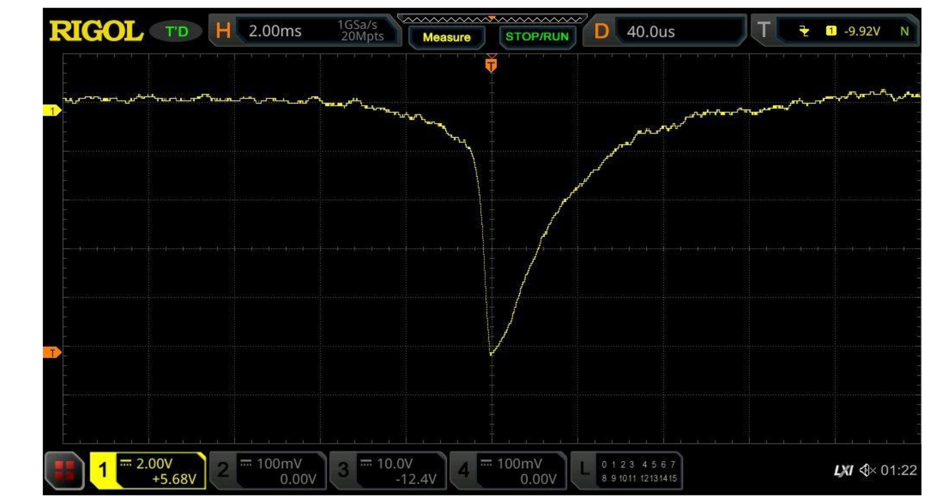
SSD Based Plasma Instrument



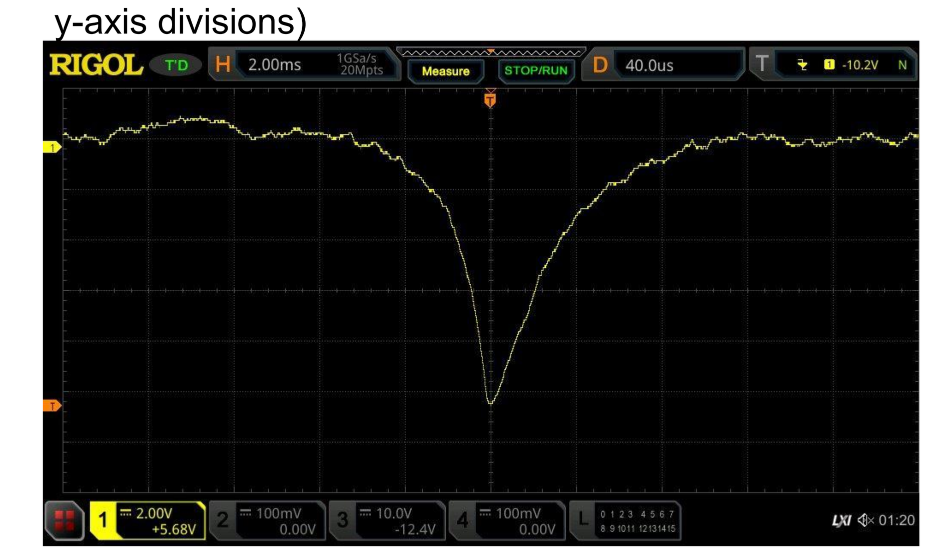
Shown here is a prototype plasma instrument under development at WVU. Hundreds of apertures operate in parallel and direct E/q resolved ions or electrons into an 8 channel SSD detector. The entire instrument is roughly 1 cm x 1 cm x 1 cm.

Proprietary SSD Results

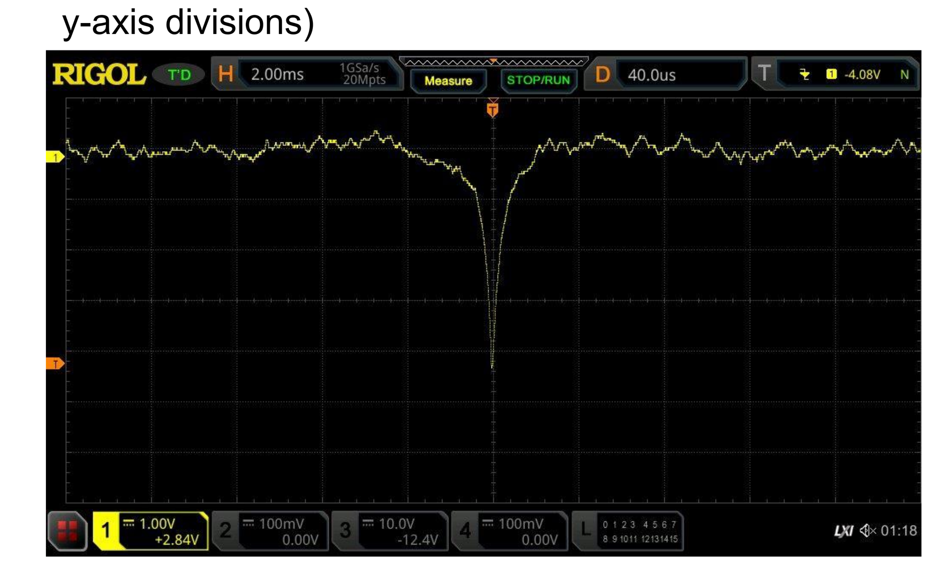
Shown here are results of a solid state, solar-blind detector being tested within SITH using the electron gun at -487.5 V bias voltage. The downward spikes indicate successful detections of electrons. The signals are amplified with a custom charge-to-voltage preamplifier.



10 keV electron detection (2ms x-axis divisions, 2 V y-axis divisions)



5 keV electron detection (2ms x-axis divisions, 2 V y-axis divisions)



1 keV electron detection (2ms x-axis divisions, 1 V y-axis divisions)

Conclusion and Future Work

- Initial tests demonstrate that the solar blind detector successfully detects electrons at energies down to at least 1 keV while remaining insensitive to visible light.
- The SITH facility is capable of testing space plasma instruments.
- The plasma chamber portion of the SITH will need more adjustments before being ready for use of plasma instrument testing.

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Characterization of an Electron Beam for Space Instrument Development

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Introduction

Getting to space is costly, time-intensive, and resource intensive. Therefore, it is important to fully calibrate scientific instruments before sending them into space. The Space Instrument Testing cHamber (SITH) is built to mimic the conditions of outer space to calibrate plasma and electric field measuring instruments.

- Using a combination of a turbo vacuum pump and cryo pump, SITH is able to reach a pressure of 8×10^{-9} Torr
- A 6-Axis motion stage on which instruments are mounted allows complete control over the orientation of the instrument with respect to the calibration electron beam.

The Space Instrument Testing cHamber (SITH)

Kimball Physics EGG 3101

Research Goal

- Discover the parameters that create a collimated beam that fully illuminates the MCP.
- Create a comprehensive look up table of electron gun parameters (electron beam current, focus voltage, *x* and *y* deflection voltages, and emission current) to obtain uniform illumination of the MCP for different electron beam energies.

Methods

A imaging microchannel plate (MCP) detector measures the number of electrons that hit the plate, as well as their position on the plate. This produces a visual map of the electron beam profile and an absolute flux measurement.

- The MCP (seen below) is mounted to the 6-Axis Stage coxially with the electron beam

The specific gun parameters that will be tested are:

- Energy
- Beam Current
- Grid Voltage
- Focus/Energy Ratio

The beam profile and total flux will be measured at various working distances, to ensure the beam is collimated, i.e., the beam is neither converging or diverging.

Acknowledgements

Sponsored in part by the West Virginia Research Challenge Fund through a grant from the Division of Science and Research, HEPC and in part by the Eberly College of Arts and Sciences

Discussion

There were a few issues involving experiment set-up that delayed data acquisition:

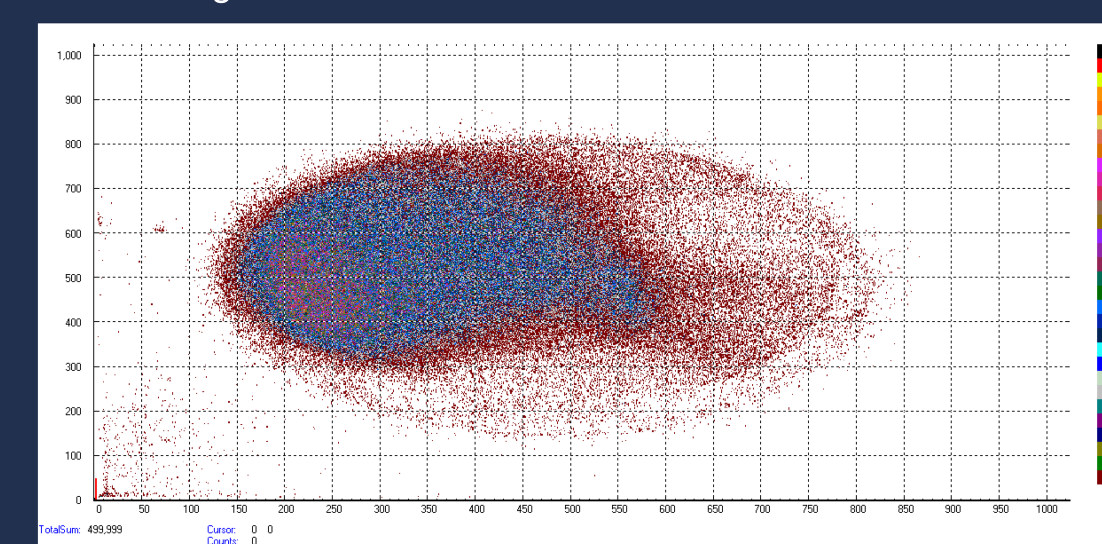
- The original 6-Axis Stage program had to be re-written in a python script to create a user-accessible interface for easy positioning
- The 6-Axis Stage, when turned on, generates interference on the MCP. However, when turned off, the stage begins to slowly fall to its lowest point.
  - A support system using constant force springs to make the stage "weightless" was designed and implemented to prevent this descent

Additional notes:

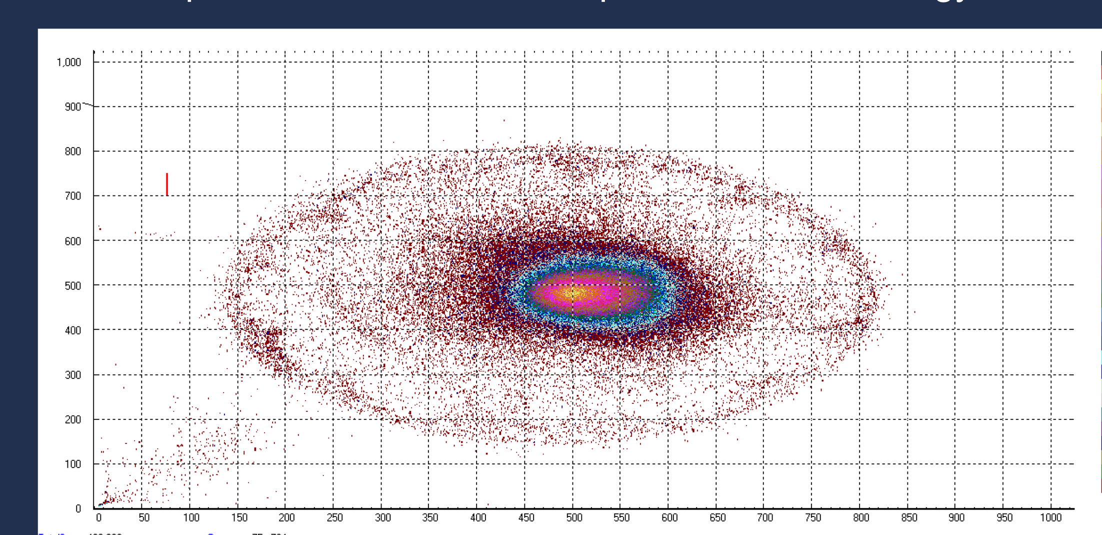
- When the MCP receives too many electron signals at once, it overloads the plate, and gives false information on the location of the electron hits, leading to faint traces (seen in the bottom left corner of each plot) of false position readings. This phenomenon is known as "pulse pileup."

Results

Both tests were performed at a beam energy of 10 keV, 0V Grid Voltage, and identical working distances; each represents 500,000 electron detections hitting the MCP.



The above plot shows the beam shape with a focus/energy ratio of 0.2.



The above plot shows the beam shape with a focus/energy ratio of 0.6, which is the recommend focus ratio for optimal beam collimation according the manual.

Conclusions and Future Work

The beam spot size is heavily dependent on the ratio of energy to focus voltage. By changing the focus voltage ratio, the beam can be collimated and maintain a constant spot size.

After the program, research on the beam characteristics will continue until the look-up table has a complete description of the full range of the electron gun's capabilities.

Once beam characterization is complete, the chamber will begin space instrument calibration for other projects.

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Helium Ion Temperatures During Magnetic Reconnection in PHASMA

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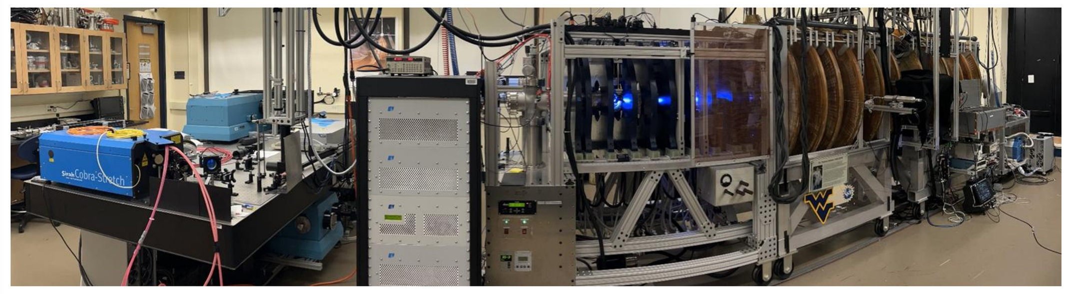
Abstract

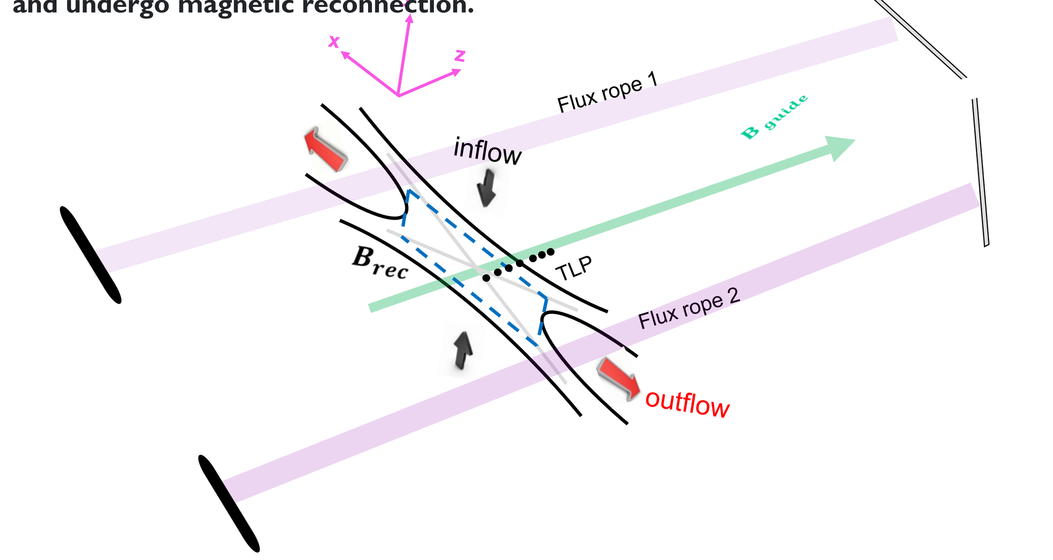
Recently, the PHASMA experiment has begun studies of magnetic reconnection in helium plasmas. Compared to argon, the smaller gyro-radius of helium allows the ions to begin to participate in the reconnection physics. However, the velocity distribution function of helium ions is not easily measured with laser induced fluorescence. To obtain a line-averaged (along the viewing chord) measurement of the helium ion temperature during reconnection, we used a custom-built LightMachinery interferometer along with an Andor ICCD camera mounted onto a 1.3m McPherson monochromator for initial measurements. The linewidth of helium ion lines is then measured as a function of time during a strong guide field magnetic reconnection study in PHASMA.

Presented here are the initial measurements of the emission lines of neutral helium made using the McPherson monochromator with a 1200 grooves/mm grating. Unfortunately, helium ion measurements using the LightMachinery interferometer were unable to be performed in time for this conference. Instead, a newly observed emission line from a Zeeman split state in helium will be shown to demonstrate the capabilities of the new spectrometer. Future studies will use this increase in precision to measure helium ion velocity distributions during magnetic reconnection.

Experimental Design



PHASMA creates two flux ropes that merge and undergo magnetic reconnection.



Because the reconnection occurs in a nearly collisionless plasma, the Hall term dominates the dissipation of energy and the change in magnetic topology during reconnection

$$\vec{E} + \vec{v} \times \vec{B} = \eta \vec{J} + \frac{1}{ne} \vec{J} \times \vec{B} - \frac{1}{ne} \vec{\nabla} \cdot \vec{P}_e + \frac{m_e}{ne^2} \frac{d\vec{J}}{dt}$$

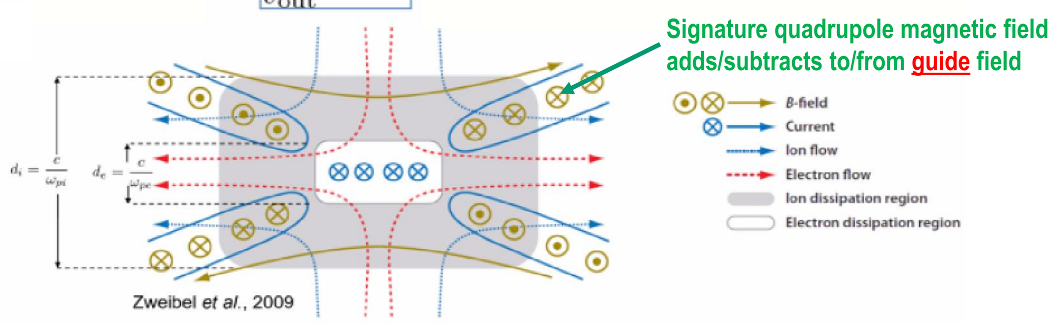
Ohm's Law

Hall Term

Electron Pressure and Inertia

$$\frac{v_{in}}{v_{out}} \sim 0.1$$

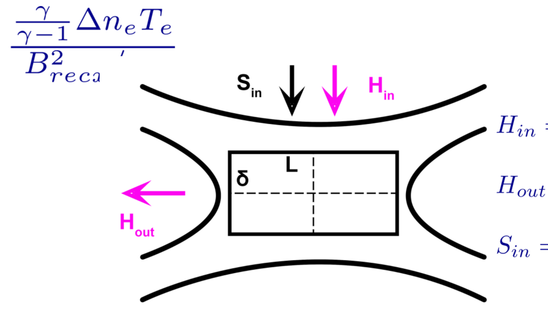
Signature quadrupole magnetic field adds/subtracts to/from guide field



The reconnection region is so small in PHASMA, and the ion gyroradius is so large (for argon), that ions are not expected to play any significant role in the reconnection process. The timescale of the entire discharge is also much faster than an ion gyroperiod.

Previous measurements of the heating of the electrons, showed that most of the available magnetic energy goes into the electrons for argon plasmas. Based on the continuity assumption, $V_{in} L = V_{out} \delta$, the ratio between electron enthalpy gain and incoming Poynting flux can be expressed as,

$$\alpha = \frac{H_{out} - H_{in}}{S_{in}} = \frac{\frac{\gamma}{\gamma-1} \Delta n_e T_e}{B^2_{recx}},$$



$$H_{in} = \frac{\gamma}{\gamma-1} n_e T_e^{in} V_{in} L$$

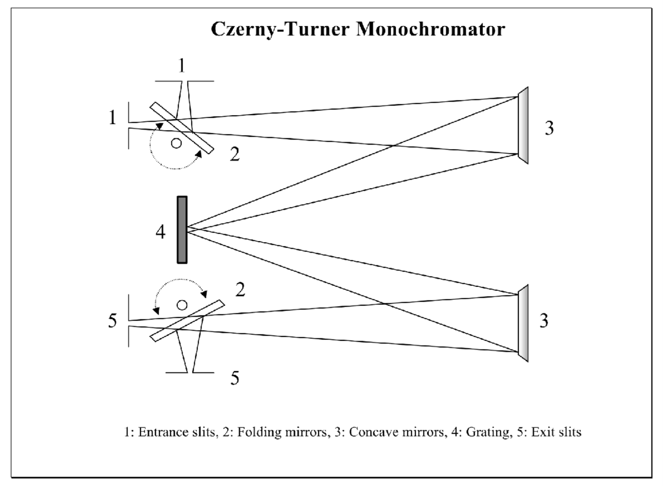
$$H_{out} = \frac{\gamma}{\gamma-1} n_e T_e^{out} V_{out} \delta$$

$$S_{in} = \frac{B^2_{recx}}{\mu_0} V_{in} L$$

T_e increases from 2.7 ± 0.1 eV to 3.0 ± 0.1 eV, corresponding to **α ~ 70%**. The question then is, if the plasma is switched to helium so that the ion gyroradius drops by a factor a ten – do ions begin to play a role in the reconnection and do they get heated?

Instruments and Apparatus Set Up

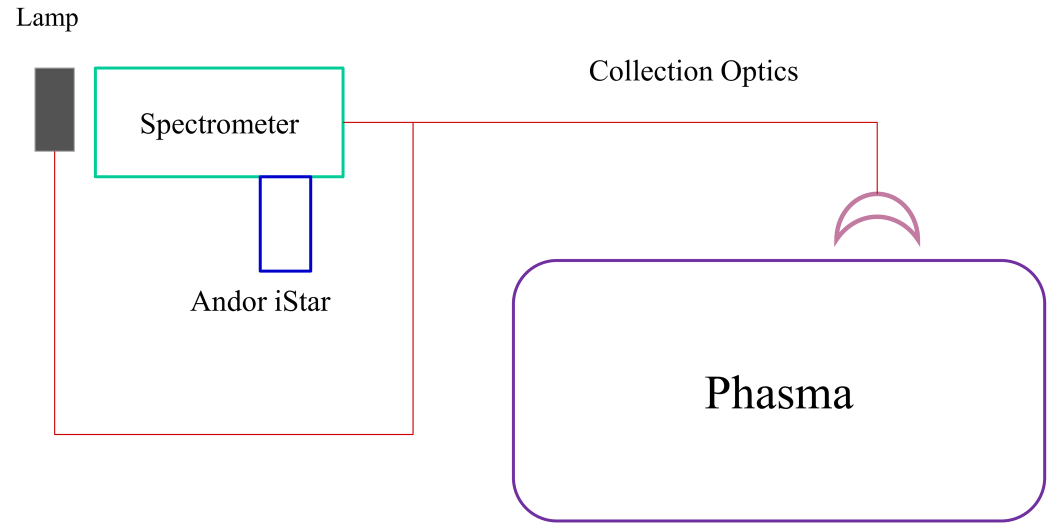
The spectrometer is a McPherson model 209 1.3 m double pass spectrometer in a Czerney-Turner configuration. Mounted on the spectrometer is an Andor iStar CCD 334 camera from Oxford Instruments. This camera is mounted to the CCD output of the spectrometer using a custom-made aluminum adapter. It was chosen due to its high resolution and picosecond gating - which enables precise control over when during magnetic reconnection the spectrum is obtained.



The collection optics looks directly into the center chamber of PHASMA where magnetic reconnection occurs and the primary diagnostics are located. Light from the experiment is coupled to the spectrometer with a 1500 micron multimode optical fiber.

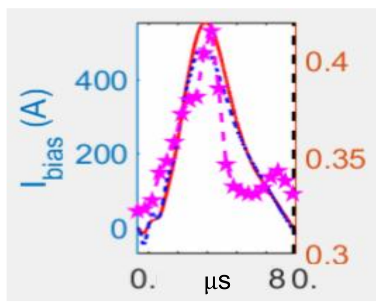
A series of spectral lamps are used to calibrate the spectrometer wavelength prior to each measurement and to provide a measurement of the instrumental resolution.

Add section about LightMachinery spectrometer

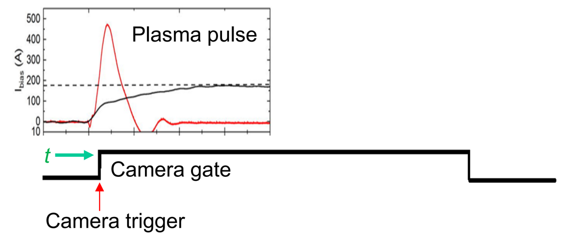


He I and Ar II Emissions

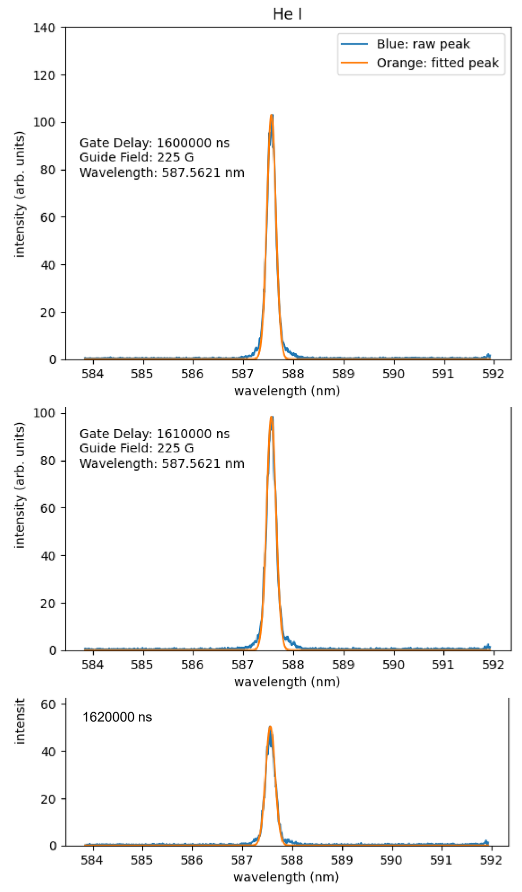
Using unnormalized measurements of the peak of the 587 nm line of neutral helium, the evolution of the helium line emission during the plasma gun pulse was measured.

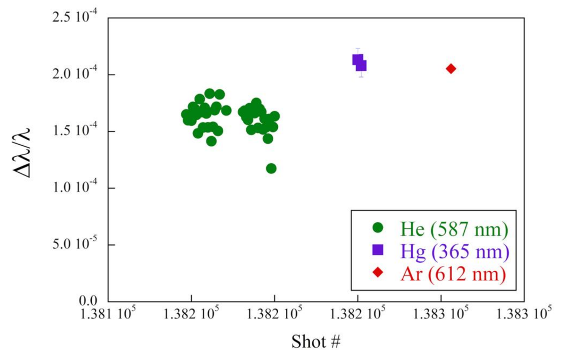


In argon plasmas, the light intensity measured with a simply photodiode often exhibits a multi-peak structure as the flux ropes undergo multiple instances of magnetic reconnection (shown at left).



The camera gate is much wider than the plasma pulse length (10 ms versus 100 µs as shown above). By stepping through the gate delays, the contribution of each interval between gate delays is estimated by subtracting the emission from the later delay from the earlier delay. The result for these helium discharges is shown below.





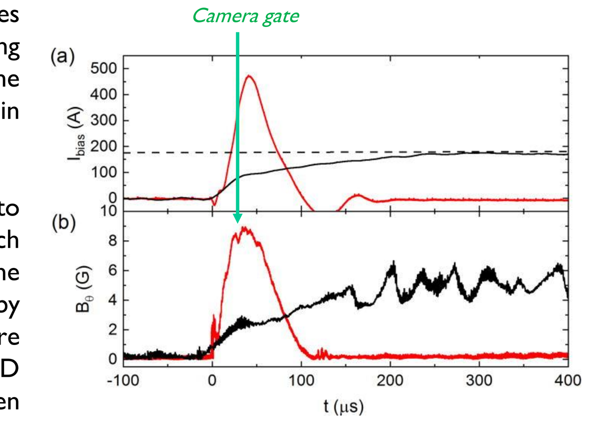
The resolution of a spectrometer depends on the input slit width used, the pixel size of the iCCD camera, and the ruling of the grating (2400 grooves/mm in this case). The intrinsic resolving power of a grating spectrometer is $\lambda/\Delta\lambda = mN$, where m is the order of the grating pattern and N is the number of grooves in the grating. Finite slit width and camera pixel size reduces the resolution from this ideal case. Therefore, the effective spectrometer resolution must be measured, in this case with the mercury lamp.

Shown above is the linewidth measured for three different cases and normalized to the center wavelength: the mercury lamp, helium neutral plasma emission and argon ion plasma emission. The linewidths are obtained from Gaussian fits to the measured lineshapes.

However, in this case, the normalized instrumental broadening (as given by the mercury lamp measurements) is comparable to the measured normalized linewidths for neutral helium and argon ions. It is possible to estimate an upper bound on the helium neutral and argon ion temperature by assuming it would be possible to detect a 1% difference in the normalized plasma emission linewidth of 0.2 x 10^{-4} for either helium or argon (a very ambitious assumption). From the Doppler broadening relations given below, a 1% increase in the linewidth corresponds to a helium neutral temperature of 0.6 eV and an argon ion temperature of 6 eV.

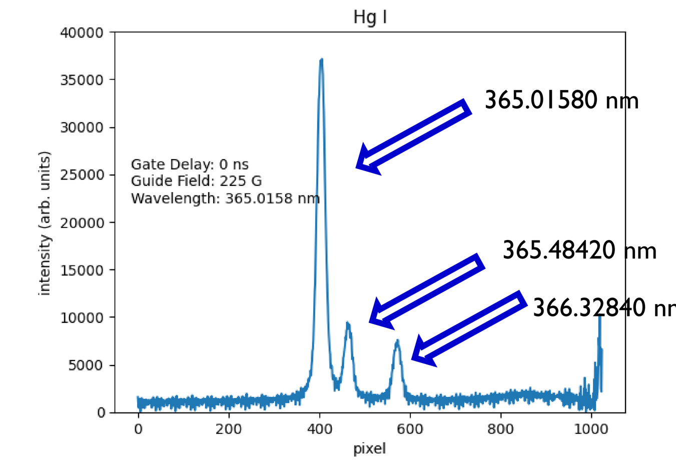
Analysis Methodology

Andor iStar camera data was exported to ASCII files along with the camera configuration, which preserves the gate delay used to trigger the camera. By stepping through gate delays it is possible to determine line width versus discharge time as well as the evolution in emission line intensity.



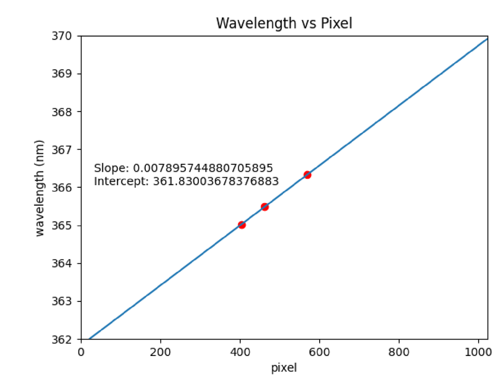
2-D camera data was imported into a Pandas library to create a database out of the raw data for each experimental configuration. Each measurement in the database was normalized by dividing every element by the max-valued element and then the columns were averaged together to generate a 1-D array. The 1-D array was low pass filtered with the SciPy filter and then the minimum value subtracted from every element to eliminate the background level.

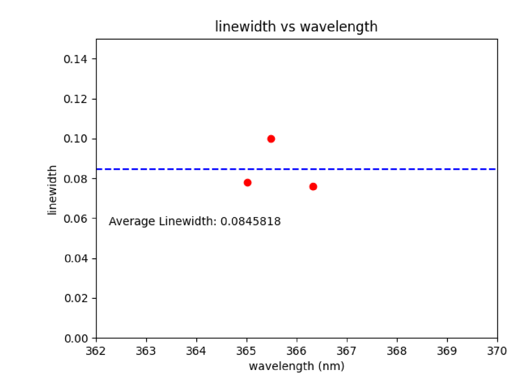
The x-axis was then converted into wavelengths from pixel value with a linear regression based on the known mercury lines in the images of the mercury lamp spectrum. Shown below is a typical mercury lamp spectrum showing three emission lines around a wavelength of 365 nm.



The linear regression for the three mercury lines is shown below along with the fit results

From the linewidths of each of the three mercury emission peaks it is possible to establish a value for the intrinsic instrumental broadening of the spectrometer-camera system. Each peak was fit to a Gaussian function to determine the linewidth.





Helium Lamp Results

Conclusion and Future Work

From the results above, we have concluded that we cannot see the change in linewidth in neutral helium due to not having enough resolution in the 1.3 m spectrometer. We have also noticed that the emission pattern for when we were looking at helium and argon ion emission lines are very similar. This implies that we are not seeing helium or argon, but instead are seeing the iron guns.

In the future, we will be getting a 2 m spectrometer, which will grant us much higher resolution to be able to see the change in the linewidth of neutral helium.

Furthermore, the shots of neutral helium will then be graphed as intensity versus time in order for us to see the evolution of the helium linewidth during magnetic recognition. From there we can pick a time to vary across different magnetic field strengths to see how that affects the linewidth of the helium.

For helium ions, we could not see the 320 nm transition line due its rarity, as such we decided to look at the 656 nm emission line. There we saw three emission lines, which we decided to compare with the same wavelength but with argon, which gave the same three emission lines. Because of this, we have determined that we are not looking at helium or argon ions, but instead we are looking at light coming from the iron and molybdenum from the ion guns. This is also supported due to the very similar linewidths.

This work is supported by the US. Department of Energy FES Award DE-SC0023040